

2nd Exam 2025/2026

Save the tasks in the file `naloge.py`.

Task 1: Read the data

The purchase data are stored in a `csv` file. Assume that the file has the format shown in the example below. A comma (,) is used as the delimiter.

```
starost, spol, izdelek, kategorija, cena, lokacija, velikost, barva, sezona, ocena,
↪  popust, placilo, pogostost
59, Female, Coat, Outerwear, 75, Texas, M, Black, Summer, 4.8, No, PayPal, Quarterly
60, Male, Shirt, Clothing, 76, Rhode Island, XL, Teal, Spring, 4.7, No, Cash, Annually
31, Female, Socks, Clothing, 43, Utah, M, Silver, Fall, 3.6, No, PayPal, Quarterly
40, Male, Hat, Accessories, 45, Iowa, M, Silver, Fall, 4.4, No, PayPal, Fortnightly
61, Female, T-shirt, Clothing, 57, Vermont, M, Gray, Summer, 2.8, No, Cash, Weekly
#63, Female, Skirt, Clothing, 90, Oregon, L, Cyan, Spring, 2.9, No, Venmo, Every 3 Months
```

Write a function `preberi_podatke` that takes the file name as an argument and returns a list of tuples of the form `(spol, kategorija, izdelek, velikost, cena, ocena)`. Here:

- `spol` represents the customer's gender, written as 'M' or 'F',
- `kategorija` represents the product category, written as `str`,
- `izdelek` represents the product name, written as `str`,
- `velikost` represents the product size, written as `str`,
- `cena` represents the product price, written as `int`, and
- `ocena` represents the product rating, written as `float`.

Determine the value of the variable `spol` by converting the gender to 'M' if the value in the file is 'Male', and to 'F' if the value in the file is 'Female'.

The file may also contain lines that start with the character `#` — skip such lines when processing the data. Also skip the line containing the column names.

Example of correct function execution:

```
>>> preberi_podatke('data/shop5.csv')
[('F', 'Outerwear', 'Coat', 'M', 75, 4.8),
 ('M', 'Clothing', 'Shirt', 'XL', 76, 4.7),
 ('F', 'Clothing', 'Socks', 'M', 43, 3.6),
 ('M', 'Accessories', 'Hat', 'M', 45, 4.4),
 ('F', 'Clothing', 'T-shirt', 'M', 57, 2.8)]
```

Task 2: Preparing a dictionary

Write a function `pripravi_slovar` that takes the following arguments:

- a list of tuples as returned by Task 1 (i.e., a list of tuples of the form `(spol, kategorija, izdelek, velikost, cena, ocena)`),
- the index of the tuple element that represents the dictionary key,
- the index of the tuple element that represents the dictionary value.

The function should return a dictionary where the keys are the values of the elements at the position given by the second argument, and the values are lists of elements at the position given by the third argument.

If the given list of tuples is empty, the function should **return** the string `"Slovarja ni mogoce pripraviti."`

If any of the given indices is invalid, the function should **return** the string `"Neveljaven indeks."`

Example of correct function execution:

```
>>> podatki = [('F', 'Outerwear', 'Coat', 'M', 75, 4.8),
               ('M', 'Clothing', 'Shirt', 'XL', 76, 4.7),
               ('F', 'Clothing', 'Socks', 'M', 43, 3.6),
               ('M', 'Accessories', 'Hat', 'M', 45, 4.4),
               ('F', 'Clothing', 'T-shirt', 'M', 57, 2.8)]
>>> pripravi_slovar(podatki, 1, 2)
{'Outerwear': ['Coat'],
 'Clothing': ['Shirt', 'Socks', 'T-shirt'],
 'Accessories': ['Hat']}
```

Task 3: Most values

Write a function `najvec_vrednosti` that takes a dictionary as an argument, where the keys are arbitrary values and the values are lists of elements.

The function should return a tuple of the form `(kljuc, vrednost)`, where `kljuc` represents the key whose value contains the largest number of elements in the dictionary, and `vrednost` is the corresponding value.

If the dictionary is empty, the function should **return None**. Be careful not to return the string `'None'`.

You may assume that exactly one key in the dictionary contains the largest number of values.

Example of correct function execution:

```
>>> slovar = {'Outerwear': ['Coat'],
              'Clothing': ['Shirt', 'Socks', 'T-shirt'],
              'Accessories': ['Hat']}
>>> najvec_vrednosti(slovar)
('Clothing', ['Shirt', 'Socks', 'T-shirt'])
```

Task 4: By gender

Write a function `po_spolih` that takes a list of tuples as returned by Task 1 (i.e., a list of tuples of the form `(spol, kategorija, izdelek, velikost, cena, ocena)`) and an optional argument `nacin`, whose default value is `"skupno"`.

Depending on the value of the argument `nacin`, the function should return:

- the set of products purchased exclusively by women (set difference) when `nacin = 'f'`,
- the set of products purchased exclusively by men (set difference) when `nacin = 'm'`,
- otherwise, the set of products purchased by both women and men (intersection, `nacin = 'skupno'`).

Example of correct function execution:

```
>>> podatki = [('M', 'Footwear', 'Sneakers', 'M', 20, 2.9),
               ('F', 'Clothing', 'Socks', 'L', 41, 4.3),
               ('M', 'Clothing', 'Sweater', 'L', 85, 3.7),
               ('M', 'Accessories', 'Gloves', 'M', 71, 3.2),
               ('M', 'Clothing', 'T-shirt', 'L', 31, 4.8),
               ('F', 'Clothing', 'Sweater', 'M', 26, 4.8)]
>>> po_spolih(podatki, nacin = 'skupno')
{'Sweater'}
>>> po_spolih(podatki, nacin = 'f')
{'Socks'}
>>> po_spolih(podatki, nacin = 'm')
{'Gloves', 'Sneakers', 'T-shirt'}
```